# Deep Reinforcement Learning for RIS-Assisted UAV Cognitive Radio with Integrated Sensing: A Reproducible Study with Optimality Certificates

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***Abstract*—A UAV secondary transmitter shares the band of a primary user (PU) under an interference limit, with the help of a reconfigurable intelligent surface (RIS). We ask whether a TD3 agent can configure the RIS in real time close to an optimisation method, and what localising a ground target costs in rate. (i) With $W = P\mathbf{v}\mathbf{v}^H$, the joint phase/power problem is exactly a rank-constrained SDP, and its relaxation was rank-one on 99.75% of the 4000 test decisions (worst randomisation gap 0.068 bit, mean $4 \times 10^{-5}$ bit): the SDR is *certified* near-optimal, which turns every comparison into an optimality gap. The alternating power/phase scheme often used instead stalls at its starting power whenever the interference constraint binds. (ii) The specified action encoding $\theta = \pi(u + 1)$ saturates the actor, while a unit-circle head learns. Even so, within our budgets TD3 reaches only 90.0% of the optimum, below closed-form co-phasing (94.9%). A residual policy anchored on co-phasing reaches 93.2%. Element-wise refinement reaches 99.7% in 0.445 ms per decision. For this problem, learning does not pay off: its speed advantage (0.0506 ms) is dominated by an analytic heuristic that is both faster and better. (iii) With an 8-antenna UAV, restricting sensing probes to the PU null space halves the median localisation error at a small sensing fraction. All numbers are regenerated from code, and an independent Octave implementation matches the simulator to $10^{-15}$.**

## I. Introduction

RIS-assisted spectrum sharing is attractive because a passive surface can simultaneously strengthen the secondary link and steer energy away from the primary receiver [1]. When the transmitter is a UAV [5], channels change continuously and the RIS must be re-configured at the control rate, which motivates learning-based controllers [2] whose inference cost is a single forward pass. We study this setting end to end, with an emphasis on *what is actually gained* by learning when strong model-based alternatives exist. Our contributions are:

- an exact joint SDP reformulation of rate maximisation under a PU constraint, a certificate of optimality from its relaxation gap, and a counterexample mechanism showing why alternating power/phase updates stall;
- an operating-regime analysis that chooses the geometry with data rather than intuition;
- a TD3 [3] design study (why the naive phase encoding fails, a circular action head, analytic power projection that makes every action feasible, a residual policy anchored on co-phasing) and an honest negative result: at this problem size, model structure beats learning;
- an ISAC time-sharing study with MUSIC, the Cramér–Rao bound and PU-aware probing [6].

## II. System Model

A single-antenna UAV flies a circle of radius $30\,\mathrm{m}$ at $100\,\mathrm{m}$ altitude and $5\,\mathrm{m/s}$. It serves a ground secondary user (SU) at $(65, 5, 0)$ through an $N$-element RIS at $(60, 0, 10)$, while a PU at $(55, -10, 0)$ tolerates interference up to $I_{\mathrm{th}}$. With cascaded channels $\mathbf{a} = \mathbf{h}_r \odot \mathbf{g}$, $\mathbf{b} = \mathbf{q}_r \odot \mathbf{g}$ and reflection vector $\mathbf{e} = [e^{j\theta_1}, \ldots, e^{j\theta_N}]^T$, we have, in noise-normalised units,

$$R = \log_2\big(1 + P|h_d + \mathbf{a}^T\mathbf{e}|^2\big), \qquad I = P|q_d + \mathbf{b}^T\mathbf{e}|^2. \quad (1)$$

Problem (P1) maximises $R$ over $(\boldsymbol{\theta}, P)$ subject to $I \le I_{\mathrm{th}}$ and $0 \le P \le P_{\max}$. Path loss is $C_0 d^{-\alpha}$ with $C_0 = -30\,\mathrm{dB}$, which fixes $\lambda = 4\pi\sqrt{C_0} = 0.397\,\mathrm{m}$. Links are Rician, and line-of-sight terms include the propagation phase. The RIS is a UPA in the $y$–$z$ plane described with direction cosines. Links with a moving end follow a Gauss–Markov model with $\rho = 0.9$. The control period is chosen so that $J_0(2\pi f_D \Delta t) = \rho$, giving $\Delta t = 8.1\,\mathrm{ms}$. Links between static nodes are block-faded.

## III. Methods

### A. Analytic power projection

Since $R$ increases with $P$ and the interference constraint is linear in $P$, the optimal power for fixed phases is

$$P^\star(\boldsymbol{\theta}) = \min\big(P_{\max},\ I_{\mathrm{th}}/|q_d + \mathbf{b}^T\mathbf{e}|^2\big). \quad (2)$$

Every method below chooses phases and uses $P^\star$, so all are feasible by construction and differ only through $\boldsymbol{\theta}$.

### B. Joint semidefinite relaxation

Let $\mathbf{v} = [\mathbf{e}; 1]$ and write $|h_d + \mathbf{a}^T\mathbf{e}|^2 = \mathbf{v}^H\mathbf{R}_{SU}\mathbf{v}$ (and similarly $\mathbf{R}_{PU}$). With $\mathbf{W} = P\mathbf{v}\mathbf{v}^H$, every diagonal entry of $\mathbf{W}$ equals $P$ because $|v_n| = 1$. Hence (P1) is *exactly*

$$\max_{\mathbf{W}\succeq 0} \ \mathrm{tr}(\mathbf{R}_{SU}\mathbf{W})$$
$$\text{s.t. } \mathrm{tr}(\mathbf{R}_{PU}\mathbf{W}) \le I_{\mathrm{th}}, \quad W_{nn} = W_{N+1,N+1} \le P_{\max}, \quad (3)$$

together with $\mathrm{rank}(\mathbf{W}) = 1$. Dropping the rank constraint gives one convex SDP [4], whose value upper-bounds (P1). Candidates are the principal eigenvector plus 100 Gaussian randomisations, each followed by $P^\star$. The alternative procedure solves an SDR at fixed $P$ and then updates $P$. When the constraint is active, the fixed-$P$ solution exhausts the interference budget, so the power update returns the same $P$: the starting point is a fixed point. Starting from $P = P_{\max}$ can also be infeasible.

### C. Element-wise refinement

Coordinate ascent over 16 phase levels on the objective $P^\star|h_{\text{eff}}|^2$, initialised at co-phasing $\theta_n = \arg h_d - \arg a_n$. Moves are accepted only if they strictly improve the objective, so the method is monotone and never worse than co-phasing.

### D. TD3 controller

The environment follows the specification: 200-step flight segments, observations containing $\mathbf{a}, \mathbf{b}, h_d, q_d$, the previous action and the last rate and interference, and reward $R$. Three design changes proved necessary. *(a) Circular action head.* With $\theta = \pi(u+1)$, the actions $u = \pm 1$ encode the same phase, but the critic extrapolates linearly past the boundary and the tanh actor saturates. We output one $(c_n, s_n)$ pair per element, normalised to the unit circle, and set $\theta_n = \operatorname{atan2}(s_n, c_n)$. *(b) Projected power.* $P = P^\star(\boldsymbol{\theta})$, which removes the penalty weight entirely. The specification's learned power with a penalty is kept as an ablation. *(c) Residual policy.* $\theta_n = \theta_n^{\text{cf}} + \operatorname{atan2}(s_n, c_n)$, so the agent learns only the PU-aware correction on top of co-phasing. Observations are phase-canonicalised by $e^{-j \arg h_d}$, which is lossless. Channel evolution does not depend on the action, so the task is a contextual bandit; this justifies the low discount $\gamma = 0.3$.

### E. Integrated sensing (Phase 4)

The UAV has an $M = 8$ ULA and splits each frame of $L$ slots, sending $\rho L$ probing slots and $(1-\rho)L$ data slots. *Communication.* Maximum-ratio transmission (MRT) reduces the problem to the single-antenna form, the element-wise method sets the RIS, and the beam is then chosen on the MRT–zero-forcing segment with $P^\star$. *Sensing.* Probes are either isotropic or confined to the PU null space, and both obey the interference limit. The echo follows the monostatic radar equation, $|\beta|^2 = C_r d^{-4}$ with $C_r = \lambda^2 \sigma_{\text{RCS}}/(4\pi)^3$. The angle is estimated with MUSIC on a $0.1^\circ$ grid [7] and compared with the deterministic CRB for one source [8]. The angle is converted to a ground position using the known range and the known target sector.

## IV. Experimental Protocol

Test results use 20 frozen flights of 200 decisions each (seeds 100–104), identical for every method. Model selection uses separate validation flights, and training flights are disjoint by construction. The Octave code reproduces the rates of the exported realisations to $1.8 \times 10^{-15}$. An independent Octave channel generator agrees within one standard error at every $N$ (Fig. 1). Compute limits (one CPU core) led to 2 seeds $\times$ 60k steps for pure TD3, 3 seeds $\times$ 30k steps for the residual agent, and 1 seed $\times$ 20k steps per ablation. The specification's budget is 5 seeds $\times$ 200k steps. TD3 is implemented in JAX and mirrors Stable-Baselines3 defaults. Runtimes are single-thread wall-clock times per decision.

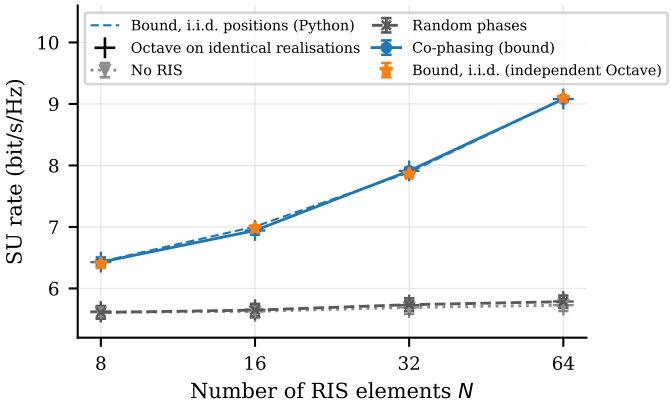



Fig. 1. SU rate vs. $N$ without the PU constraint ($P = P_{\max}$). Octave recomputation on identical realisations (+) and an independent Octave generator ($\star$) overlay the Python curves.

## V. Results

### A. Operating regime

At the specified geometry, the direct link is 7 dB stronger than the best-case RIS path at $N = 16$. Even so, the RIS is worth 1.49 bit/s/Hz, because the PU constraint binds in 81.5% of states and the RIS relaxes it. PU-aware configuration adds 0.27 bit/s/Hz over co-phasing. With 10 dB of extra direct-link blockage, the constraint binds in only 7.5% of states and the PU-aware gain collapses to 0.02. We therefore keep the specified geometry.

### B. Optimisation baselines

Without the constraint, co-phasing grows from 6.43 to 9.08 bit/s/Hz between $N = 8$ and 64, while random phases stay at the no-RIS level (Fig. 1). Under the constraint (Figs. 2–3), all methods saturate once $P_{\max}$ exceeds the interference-limited power. Above that point, the gap between PU-agnostic co-phasing and the PU-aware methods is what the RIS buys by steering energy away from the PU. The element-wise heuristic nearly closes this gap on average (99.7% of the optimum on the test flights). Its worst decision still loses 0.40 bit, and it is within 0.01 bit of the optimum on only 64% of decisions. The joint SDR meets its own upper bound at every sweep point.

### C. Learning

The specified tanh encoding saturated immediately: 100% of actions ended up at $|u| > 0.999$, and the validation rate stayed at the random-phase level (Fig. 4b). The circular head learns steadily but slowly. After 60k steps, pure TD3 reaches $5.21 \pm 0.001$ bit/s/Hz on the test flights, i.e. 90.0% of the certified optimum, below PU-agnostic co-phasing (5.49, 94.9%) (Table I). The residual policy starts *exactly* at co-phasing, because its untrained correction is near zero. TD3 then first degrades it, as early critic errors pull the actor away, and only slowly recovers: $5.39 \pm 0.015$ after 30k steps (93.2%). Validation-based model selection that also included the untrained policy would simply return co-phasing. The ablations confirm the design choices at equal step counts.

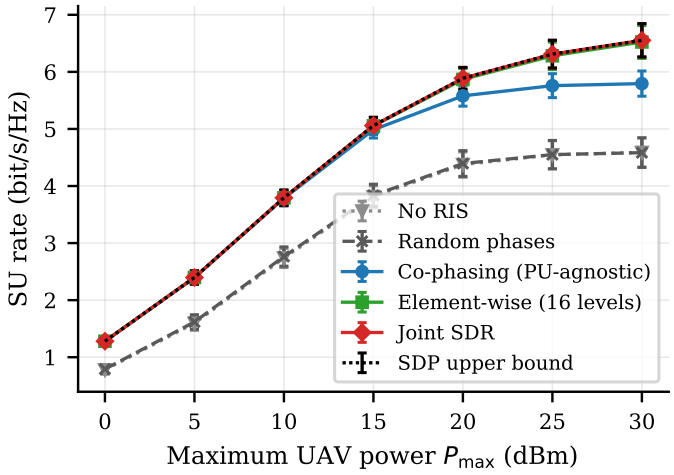



Fig. 2. SU rate vs. $P_{\max}$ ($N = 16$, $I_{\text{th}} = -90$ dBm). Error bars: 95% CI.

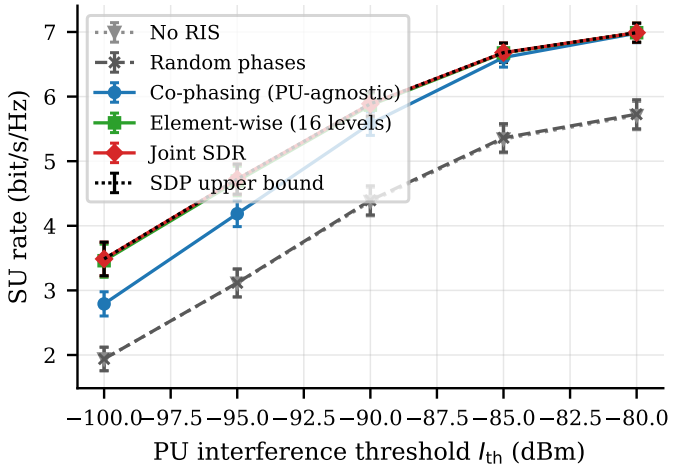



Fig. 3. SU rate vs. $I_{\text{th}}$ ($N = 16$, $P_{\max} = 20$ dBm).

$\gamma = 0.99$ learns markedly slower (4.45 bit/s/Hz on test after 20k steps), consistent with a contextual bandit, where bootstrapping only adds critic noise. With learned power, both penalties fail. The specified linear penalty still violates the PU limit in 24.6% of decisions, at a rate of only 4.30. The bounded dB penalty is simply too cheap: the agent reports 6.35 bit/s/Hz, *above* the certified optimum, by violating the limit in 78.9% of decisions. This is the textbook case of a good reward hiding violations. The projected-power agents are feasible by construction. Runtime is where learning should shine (Fig. 5): the TD3 forward pass costs 0.0506 ms at $N = 16$ and 0.0709 ms at $N = 64$, versus 139 ms and 6115 ms for the SDR (a speed-up of 86269× at $N = 64$). But co-phasing costs 0.00867 ms and element-wise 0.445 ms while being more accurate. The honest conclusion is that, for one SU and one PU, DRL is dominated. It becomes interesting only where no cheap near-optimal heuristic exists (multi-user, multi-PU, imperfect CSI), which is where we would take it next. Under one control period of CSI delay (Table I, right), every method keeps most of its rate. However, 44.3% of SDR decisions (and a similar share for every method) then exceed the PU limit. Whenever the constraint binds, projection operates exactly at the limit, so any channel change pushes the interference above it about half of the time; a back-off margin is needed in practice.

### D. Integrated sensing and communication

MUSIC attains $\sqrt{\text{CRB}}$ above its SNR threshold and then reaches the grid floor (Fig. 7). With fewer snapshots than antennas, MUSIC does not break down for a single target. Only its threshold moves, because only the principal eigenvector is needed. In the trade-off (Fig. 8), the 8-antenna UAV nulls the PU for data, reaching $R = 9.52$ bit/s/Hz. Isotropic probing, however, remains interference-limited, with a median echo SNR of -4.0 dB. Restricting probes to the PU null space raises it to -0.1 dB. At $\rho = 0.1$ this cuts the median localisation error from 2.96 to 1.45 m, and the 5 m outage from 31% to 10%, which is comparable to doubling the frame length.

## VI. Discussion and Limitations

Learning is only worth its training cost if it beats cheap model-based alternatives on the metric that matters. Here, strong structure (co-phasing, analytic power, a tight SDR) makes the element-wise heuristic nearly optimal at sub-millisecond cost, and a certified optimum makes that visible. This is the main methodological point: without the SDP bound, a TD3 agent at 95% of "SDR" could be misread as a success. Limitations: perfect current CSI of the PU cascade is assumed (one-step-old CSI is reported); RIS elements are isotropic with no wall shadowing; the RL budgets are below specification (2–3 seeds, 30–60k steps instead of 5 × 200k), although the learning curves are still rising slowly; the SDR at $N = 64$ uses SCS because the interior-point solver exceeds 3 GB; and the position error assumes a known range.

## VII. Conclusion

(P1) is an exact rank-constrained SDP whose relaxation was rank-one on 99.75% of test decisions and near-tight on the rest. Power projection and co-phasing capture most of the achievable rate, and element-wise refinement closes the rest. TD3 needs a periodic action representation even to learn. At this size, it is dominated by an analytic heuristic in both rate and latency. For ISAC under an interference limit, PU-aware probing is the lever that matters.

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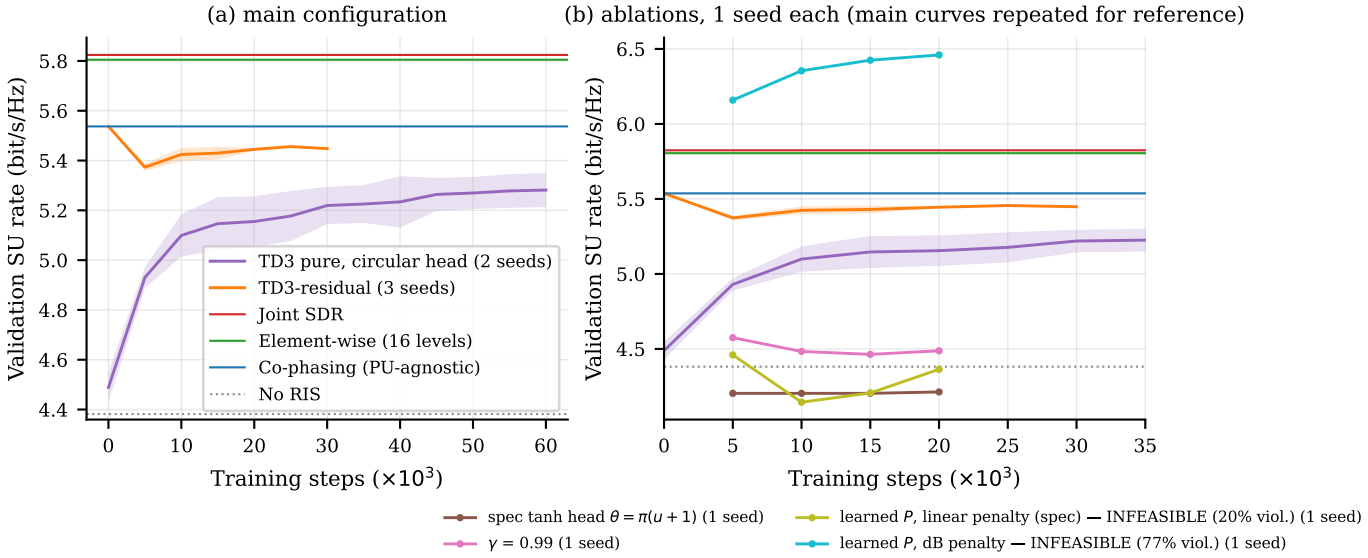



Fig. 4. Validation rate during training (mean ± std over seeds). (a) Main agents vs. baseline levels. (b) Ablations at equal step counts.

TABLE I

TEST FLIGHTS ($N = 16$, 4000 DECISIONS PER METHOD). TD3 ROWS: MEAN OVER SEEDS (± STD). TIME: IDLE-CORE BENCHMARK. RIGHT: SAME DECISIONS APPLIED ONE CONTROL PERIOD (8.1 MS) LATE.

| Method | Rate (bit/s/Hz) | % of SDR | Violations (%) | Time/decision (ms) | Rate, old CSI | Violations, old CSI (%) |
|---|---|---|---|---|---|---|
| No RIS | 4.279 | 73.9 | 0.0 | 0.00329 | 4.276 | 43.2 |
| Random phases | 4.281 | 74.0 | 0.0 | 0.00643 | 4.278 | 42.8 |
| Co-phasing (PU-agnostic) | 5.491 | 94.9 | 0.0 | 0.00867 | 5.405 | 42.9 |
| Element-wise (16 levels) | 5.767 | 99.7 | 0.0 | 0.445 | 5.675 | 44.1 |
| Joint SDR | 5.787 | 100.0 | 0.0 | 139 | 5.697 | 44.3 |
| TD3 (pure, circular head) [2 seeds] | 5.207 ± 0.001 | 90.0 | 0.0 | 0.0506 | 5.125 | 42.8 |
| TD3-residual (physics-informed) [3 seeds] | 5.395 ± 0.015 | 93.2 | 0.0 | 0.0555 | 5.310 | 42.9 |
| TD3, spec tanh head [1 seeds] | 4.295 | 74.2 | 0.0 | 0.0506 | 4.292 | 43.2 |
| TD3, gamma=0.99 [1 seeds] | 4.446 | 76.8 | 0.0 | 0.0506 | 4.415 | 42.8 |
| TD3, learned P + linear penalty [1 seeds] — INFEASIBLE | 4.301 | 74.3 | 24.6 | 0.0506 | 4.255 | 30.8 |
| TD3, learned P + dB penalty [1 seeds] — INFEASIBLE | 6.347 | 109.7 | 78.9 | 0.0506 | 6.274 | 79.0 |
| SDP upper bound | 5.787 | | | | | |

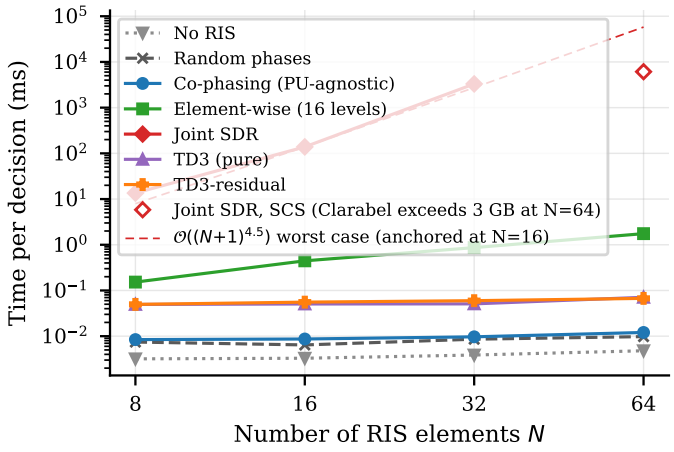



Fig. 5. Time per decision vs. $N$ (single thread, log scale). The TD3 time is architecture-bound and does not depend on the weights.

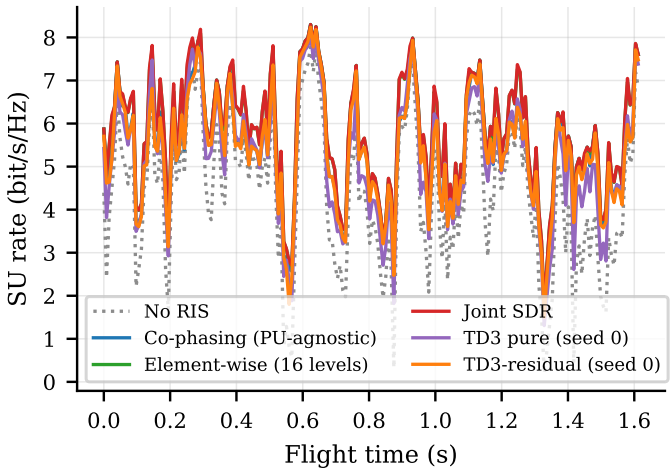



Fig. 6. SU rate along test flight 0 (1.6 s, 200 decisions).

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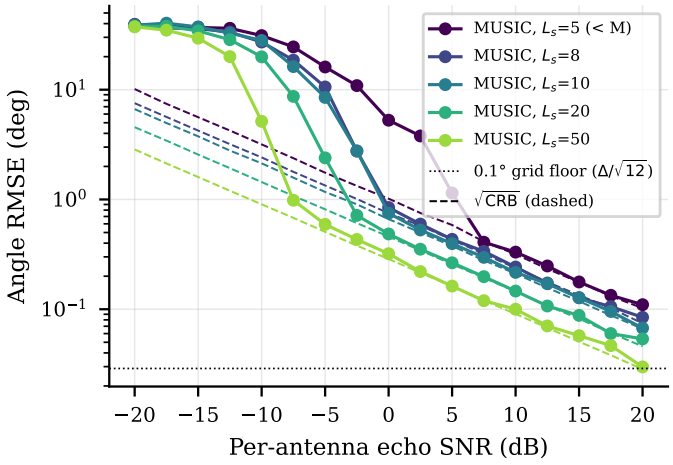



Fig. 7. Angle RMSE of MUSIC (markers) and $\sqrt{\mathrm{CRB}}$ (dashed) vs. per-antenna echo SNR, $M = 8$, 500 trials per point.

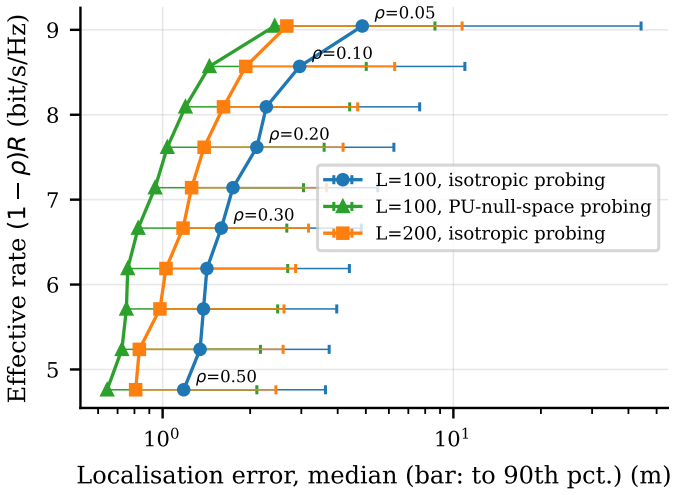



Fig. 8. Effective rate vs. localisation error as the sensing fraction $\rho$ increases (500 trials, common random numbers across $\rho$).

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